Solution Requirements

Date	27 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Bank credit officers and financial analysts securely log in using username and password to access the Smart Lender system.	High
2	Authorization levels	The system provides rolebased access. Credit Officers can evaluate loan applications, Financial Analysts can view reports, and Administrators can manage users and ML models.	High

3	External interfaces	The application interacts with the trained XGBoost model, Flask backend, IBM Cloud platform, and applicant database for prediction and storage.	High
4	Transactions processing	The system accepts applicant details, validates the data, predicts loan approval using the ML model, and stores the prediction results in the database.	High
5	Reporting	Generate loan approval reports, prediction summaries, applicant history, approval/rejection statistics, and model performance reports.	Medium
6	Business rules, etc.	Applicants with good credit history and sufficient income have higher approval probability. Loan eligibility is determined based on applicant details and the trained XGBoost model.	High
7	Compliance to laws or regulations	Ensure applicant data privacy and secure storage by following banking security guidelines and applicable data protection standards.	High

8	Other	Provide a user-friendly web interface that instantly displays loan approval or rejection after prediction. Medium	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1	Performance & Speed	The system should generate loan predictions quickly after applicant details are submitted.	`< 500ms` under normal load
2	Scalability	The application should support multiple users submitting loan applications simultaneously.	Supports 1000+ concurrent users
3	Security & Data Privacy	Applicant information must be protected using secure authentication, encrypted communication, and secure database storage.	HTTPS, encrypted data transmission, secure login
4	Reliability & Availability	The Smart Lender application should remain available for banking operations with minimal downtime.	99.5% uptime

5	Usability & Accessibility	The application should have a simple, responsive interface that is easy for applicants and banking staff to use.	Responsive design and intuitive navigation
6	Other	The system should be easy to maintain, update with new ML models, and deploy on IBM Cloud or other cloud platforms.	Modular Python-Flask architecture
7	**Model Accuracy**	The underlying classifier must have sufficient precision and recall in detecting credit default risk based on the historical training set.	>80% test accuracy